

Project

Stop-loss and Leverage in optimal Statistical Arbitrage

Financial Engineering AY 2024-2025

Consider the pair of the two futures in the energy market Heating Oil c2 (HOc2) and Gas-Oil c6 (LGOc6) in the time interval between the 22.04.2015 and 21.04.2016 (dataset in HO-LGO.xlsm). Divide the half-an-hour time series in an In-Sample (IS) period of 9 months and Out-the-Sample (OS) period of 3 months.

- a. Prove that an OU process admits always a profitable trading strategy also including a stop-loss if transaction costs are lower than a critical value; in particular focus on the case of low steady state standard deviation for the OU dynamics Σ (see e.g. [1]). Reproduce Figure 3.
- b. After having filtered out outliers (see e.g. [1]), it is required to calibrate IS the ratio of the two prices with an OU, providing also Confidence Intervals (CI) for parameter estimations via a (statistical) bootstrap technique as described in [1]. Consider only prices in the time window is 9:00-16:00 New York Time (NYT) as suggested in [1]. What is the difference in the parameters if the time window is 8:00-16:00 NYT (mention in this case the outliers that have been excluded)?
- c. Estimate transaction cost parameter as the expected value in the IS period. With a stop-loss level $l=-1.645$, compute the optimal IS bands and the optimal IS long-run return with their CI and the corresponding OS long-run return using round-the-clock data (with the exception of the time interval 17:00-20:00, when one of the two markets is closed).
- d. Reproduce the results for optimal return for different values of leverage ($f=1,2,5$, optimal), see e.g. Table 2.
- e. What are the results of point c. that can be obtained with different stop-loss levels e.g. $l=-1.282$, $l=-1.96$ and $l=-2.326$?

g[Facultative]. Repeat the analysis of points c-e for a different pair of assets in the given spreadsheet that collects close prices for 1 minute frequency in the last 6 months for governative futures (where IKA stands for next expiring future whose underlying is the 10y Italy, OATA 10y France, OEA 5y Germany and RXA 10y Germany). You can assume that bid/ask is of 1 tick (e.g. 131.08-131.09).

h[Facultative]. Repeat the analysis of points c-e for a different pair of assets at your choice, with any other hi-frequency dataset.

Realize a library in Matlab. Optional Python.

Hints:

- a. Consider time intervals as regularly spaced in the IS and OS periods. Pay attention that the two time intervals are different.

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[1] R. Baviera & T. Santagostino Baldi, (2019). Stop-loss and leverage in optimal statistical arbitrage with an application to energy market. *Energy Economics*, 79, 130-143.